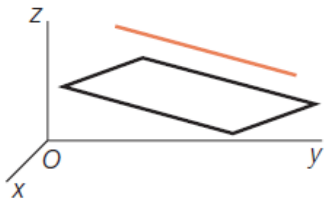
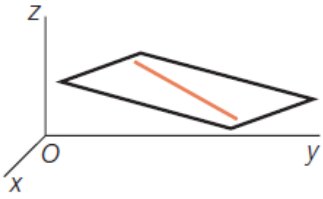
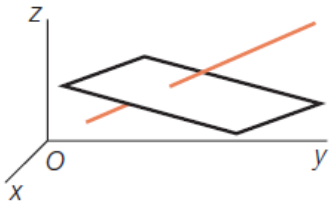
Intersection of a Line and a Plane

		
Line is _____ to the plane	Line _____ in the plane	Line _____ the plane
No Solution	Infinite # of Solution	One Solution

To determine the solution(s), write the equation of a line in _____ form and sub into the _____ equation of the plane.

Example 1: Determine the intersection of the given line and plane.

$\vec{r} = (2,3,-1) + t(2,1,-5),\ t \in \Re$ and $4x - 3y + 2z - 12 = 0$

Example 2: Determine the intersection of the given line and plane.

$$\left.\begin{aligned} x &= 1+3t \\ y &= 2+2t \\ z &= 3 \end{aligned}\right\} \; t \in \Re \qquad \text{and} \qquad 2x-3y+4z+1=0$$

Example 3: Determine the intersection of the given line and plane.

$$x-3=\frac{z+2}{2},\; y=5 \qquad \text{and} \qquad 4x+3y-2z-31=0$$